

Solution:

(1)

$$1) E_x = \int_{-\infty}^{+\infty} (\text{sinc}^2(t-3) + \text{sinc}^2(t+2) + 2 \text{sinc}(t-3) \text{sinc}(t+2)) dt$$

$$\int_{-\infty}^{+\infty} \text{sinc}^2(t-3) dt = \int_{-\infty}^{+\infty} \text{sinc}^2(t+2) dt = \int_{-\infty}^{+\infty} \text{sinc}^2 t dt = \text{triangle}(0) = 1$$

$$\int_{-\infty}^{+\infty} \text{sinc}(t-3) \text{sinc}(t+2) dt = \int_{-\infty}^{+\infty} \text{rect}(f) \cdot e^{-j2\pi 3f} \text{rect}(f) e^{-j2\pi 2f} df$$

Parseval

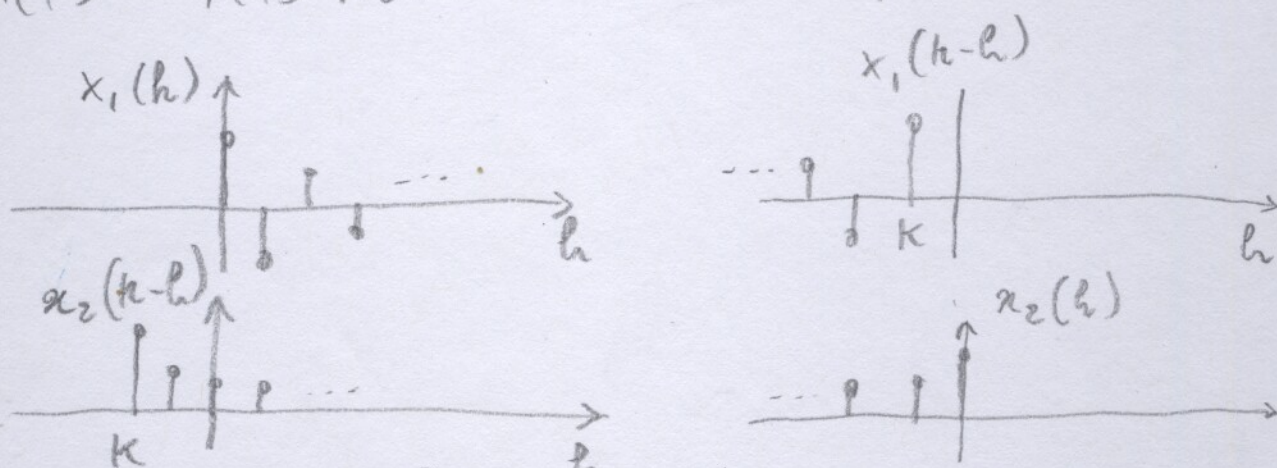
$$= \int_{-\infty}^{+\infty} \text{rect}(f) e^{-j2\pi 5f} df = \text{sinc}(-5) = 0$$

$$\boxed{E_x = 2}$$

$$2) \frac{1}{1+0.2e^{-j2\pi f}} \leftrightarrow (-0.2)^k 1_0(k) = x_1(k)$$

$$\frac{1}{1-0.1e^{+j2\pi f}} \leftrightarrow (0.1)^{-k} 1_0(-k) = x_2(k)$$

$$X(f) = X_1(f) \cdot X_2(f) \leftrightarrow x(k) = x_1 * x_2(k)$$



$$k \leq 0 \quad x(k) = \sum_{l=0}^{+\infty} (-0.2)^l (0.1)^{-(k-l)} = 0.1 \sum_{l=0}^{-k+\infty} (-0.02)^l$$

$$= (0.1)^{-k} \frac{1}{1+0.02}$$

$$k > 0$$

$$\sum_{l=-\infty}^0 (0.1)^{-l} (-0.2)^{k-l} = (-0.2)^k \frac{1}{1+0.02}$$

(2)

$$E_x = \left(\frac{1}{1.02}\right)^2 \sum_{l=-\infty}^0 (0.1)^{-2l} + \left(\frac{1}{1.02}\right)^2 \sum_{k=1}^{+\infty} (-0.2)^{2k}$$

$$\bar{E}_x = \left(\frac{1}{1.02}\right)^2 \left[\frac{1}{1-0.01} + \frac{1}{1-0.04} - 1 \right]$$

3) $b(kT)$ e $a(kT)$ sono simboli indipendenti in quanto $a(kT)$ e $b(kT)$ sono simboli indipendenti

$$P[b(kT) = 1] = P[a(kT) = 1] + P[a(kT) = 2] = 0.4$$

$$P[b(kT) = -1] = 0.6$$

$$M_b = E[b^2(kT)] = E[1] = 1$$

$$m_b = 0.4 \cdot 1 + 0.6 \cdot (-1) = -0.2$$

$$\sigma_b^2 = M_b - m_b^2 = 1 - 0.04$$

$$R_b(f) = \sigma_b^2 \cdot T + m_b^2 \delta_{\frac{1}{T}}(f) \leftarrow \text{simboli ind.}$$

4) Il processo ha banda $\frac{1}{T} = 5 \text{ MHz}$

Teorema di Shannon: $\frac{2}{T} = 10 \text{ Mcamp/s}$

Densità uniforme, $(\text{SNR})_{\text{dB}} \approx 6 \text{ m} \Rightarrow m = 8 \text{ bit/campione}$
 per avere $(\text{SNR})_{\text{dB}} > 45$

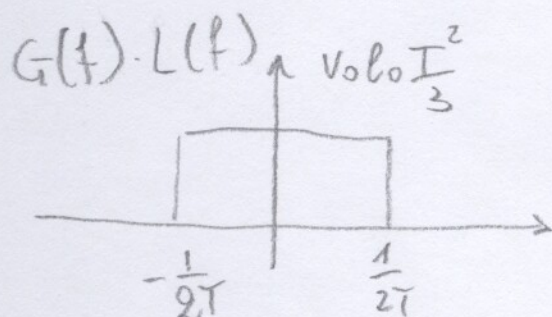
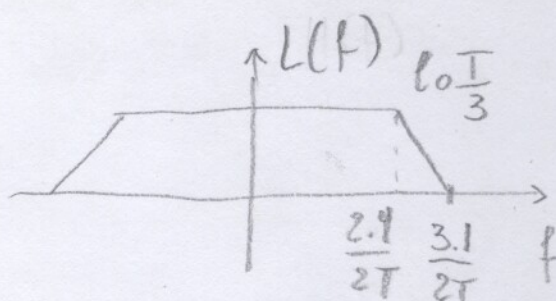
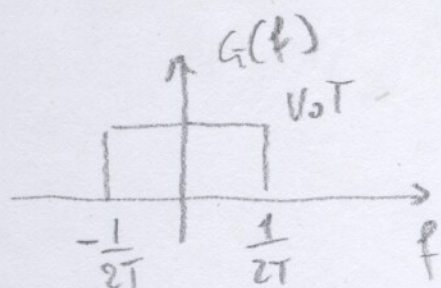
Flusso: 80 Mbit/s . 64-QAM $\Rightarrow 6 \text{ bit/simbolo}$

Occorre trasmettere $\frac{80}{6} \text{ Msimboli/s}$. Banda minima

richiesta: $\frac{80}{6} \text{ MHz}$. Con roll-off 0.25, Banda

$$\text{richiesta } B_v = 1.25 \cdot \frac{80}{6} \text{ MHz}$$

5)



$C(f) = G(f)L(f)H(f)$ la banda massima $\frac{1}{2T}$. Per avere $C(f)$ di Nyquist, deve essere $C(f) = V_0' T \text{ rect}(fT)$, e dunque

$H(f) = k \text{ rect}(fT)$ (in modo da limitare il rumore)

$$C(f) = k V_0 T l_0 \frac{I}{3} \cdot \text{rect}(fT) \quad c(t) = \frac{k V_0 T l_0}{3} \text{vnc}\left(\frac{t}{T}\right)$$

$$V_0' = \frac{k V_0 T l_0}{3}$$

$$\sigma_n^2 = R_0 E_n = \frac{R_0 k^2}{T}$$

$$P_c = P[q_k = 1, n_k > \frac{V_0'}{2}] + P[q_k = 0, -\frac{V_0'}{2} < n_k < \frac{V_0'}{2}] + P[q_k = -1, n_k < -\frac{V_0'}{2}]$$

$$= \frac{1}{4} \phi\left(\frac{V_0'}{2\sigma_n}\right) + \frac{1}{2} \left[\phi\left(\frac{V_0'}{2\sigma_n}\right) - \phi\left(-\frac{V_0'}{2\sigma_n}\right) \right] + \frac{1}{4} \phi\left(\frac{V_0'}{2\sigma_n}\right)$$

$$P_e = 1 - P_c = \frac{3}{2} Q\left(\frac{V_0'}{2\sigma_n}\right)$$